InterviewGen AI

AI-Based Mock Interview and Performance Analyzer

Project Documentation

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Team ID: Byte-Bloom

Team GitHub Repository: https://github.com/KANIKARAMPOOJITHA/gen-ai-project-InterviewGen-AI-AI-Based-Mock-Interview-Performance-Analyzer.git

Student Name	Registration Number	Individual GitHub Link
KANIKARAM MEDHARA POOJITHA	99230040116	github.com/KANIKARAMPOOJITHA
CHEEMALA SIVA MOWNY	99230040092	github.com/cheemalasivamowny
GAJULA HEMALATHA	99230040300	github.com/99230040300
GANDHAM PRAVALLIKA	99230040302	github.com/pravalikagandham4

Project Trainer / Guide

Mr. SHIRWIN PRINCE I

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1.About this document

This document explains the purpose, main features, and intended workflow of InterviewGen AI in straightforward language. It is based on the project dashboard template provided. Where the supplied material does not show a technical detail or verified test result, the document leaves that detail open rather than guessing.

2.Introduction

Preparing for an interview is not always easy, especially when a student is unsure what to practise, how to explain an answer, or how to judge whether they are improving. InterviewGen AI was developed to bring several parts of interview preparation together in one place.

The project is designed as an interview-preparation workspace. It brings together mock interview practice, resume analysis, an AI interview assistant, voice practice, performance analytics, and an improvement plan. Instead of treating preparation as a one-time activity, the platform encourages users to practise, review their performance, and work on areas that need attention.

The aim is practical: help users feel more prepared by giving them a place to practise and reflect before attending a real interview.

3.Problem Statement

Students often prepare for interviews using separate websites, notes, videos, and practice sessions. This can make preparation feel scattered. It may also be difficult to get useful feedback after answering a question or to keep track of progress over time.

InterviewGen AI addresses this problem by providing a single platform for different interview-preparation activities. It is intended to make practice easier to access and help users identify what they should focus on next.

4.Objectives

The main objective is to build a user-friendly platform that supports a student's interview preparation from practice through review.

The specific objectives are to provide a mock interview area for technical, HR, and behavioural practice; offer resume analysis; provide an AI assistant for interview and technical-learning questions; support spoken-answer practice; present performance information; and guide users towards focused improvement.

The project also aims to keep the dashboard simple to navigate, so users can move between the available tools without needing to learn a complicated interface.

5.Overview of the System or System architecture

InterviewGen AI is organised around a dashboard. The dashboard acts as the starting point and gives users access to the main preparation tools. It also presents summary areas for interview activity, scores, resume analysis, practice time, progress, skill readiness, certificates, and grade points.

The dashboard includes links to the Mock Interview, Resume Analyzer, AI Interview Assistant, Voice Practice, Analytics, and Certificates sections. It also includes an Improvement Plan area with guidance grouped into strengths, areas to improve, AI recommendations, and weekly goals.

Some dashboard values are displayed as empty or default values until a user has completed the relevant activity. This is important because progress figures should reflect actual user activity rather than being presented as completed results when no practice has taken place.

6.Main Features

Mock Interview: This section is intended to let users practise interview questions in areas such as technical knowledge, HR, and behavioural topics. Users can use it to build familiarity with interview-style questions and practise giving structured answers.

Resume Analyzer: This feature allows a user to submit a resume for review. The dashboard describes the purpose as identifying skills, strengths, missing areas, and topics that may need further preparation.

AI Interview Assistant: The chatbot provides a place to ask questions about topics such as coding, Python, Java, SQL, AI/ML, HR interviews, and placement preparation. It is intended to support learning and clear up doubts during preparation.

Voice Practice: This area focuses on spoken answers and communication. The dashboard describes possible feedback areas such as filler words, speaking rate, fluency, and communication quality.

Performance Reports and Analytics: These sections are intended to help users review interview scores, strengths, weaknesses, skill development, and preparation consistency over time.

Improvement Plan: The dashboard groups preparation guidance into strengths, areas to improve, AI recommendations, and weekly goals. The intention is to turn feedback into manageable next steps.

Certificates and Grade Points: The dashboard includes an achievements section where users can view certificates earned and accumulated grade points. These values are expected to depend on completed interview sessions and the project's achievement rules.

7.Technology and Implementation

The supplied dashboard template uses HTML for page structure, CSS for styling, JavaScript for front-end behaviour, and Flask/Jinja-style template expressions for dynamic links and values. For example, route helpers are used to connect dashboard links to application pages, while template variables are used to display certificate and grade-point totals.

The dashboard is designed with a sidebar, a top navigation area, summary cards, feature cards, progress panels, and a footer. Its layout uses responsive CSS so that key sections can rearrange for narrower screens. The visual design uses a dark interface with purple accents to give the platform a consistent appearance.

The complete backend, database structure, AI model or API provider, authentication flow, and deployment configuration were not included in the dashboard template supplied for this documentation. Those details should be added from the actual application source code rather than assumed.

8.Typical User Workflow

A user begins at the dashboard and chooses the activity they want to work on. They might start a mock interview, upload a resume for analysis, ask the AI assistant a question, or open voice practice. After completing practice activities, the user can visit the analytics and reports sections to review available feedback.

The user can then refer to the Improvement Plan and choose a practical next step, such as revising a weak topic or completing another mock interview. Over time, repeating this cycle—practise, review, and improve—is expected to make preparation more organise.

9.Implementation Details & Tech Stack

Component	Technology Used	Purpose
Frontend	HTML5	Creates the structure of the web application.
Styling	CSS3	Designs the layout, colors, buttons, cards, and responsive interface.
Client-Side Scripting	JavaScript	Handles user interactions and dynamic frontend behavior.
Backend Framework	Python Flask	Manages application routes and server-side operations.
Template Engine	Jinja2	Displays dynamic content and connects backend data with HTML pages.
AI Integration	AI service/API – to be confirmed	Supports AI-based interview questions, feedback, and analysis.
Resume Analysis	To be confirmed	Intended to evaluate uploaded resumes and provide feedback.
Voice Processing	To be confirmed	Intended to support voice-based interview practice.
Database	To be confirmed	Intended to store user details, interview results, scores, and progress.
Development Environment	To be confirmed	Used to develop, test, and run the application.

10.Benefits

The main benefit of InterviewGen AI is that it brings several preparation activities into one workspace. Users do not have to think of mock interviews, resume review, communication practice, and progress tracking as completely separate tasks.

The platform also encourages regular practice. By making feedback and improvement guidance part of the same experience, it gives users a clearer way to decide what to work on after each session.

11.Testing and Evaluation

The dashboard template shows the intended interface and navigation for the main features. However, the template alone does not prove that every feature, AI response, score calculation, or report has been fully tested.

Before final submission, each feature should be tested using realistic inputs. Navigation links should be checked, invalid or empty inputs should be handled clearly, and any displayed scores or achievements should be compared with the actual stored results. Performance claims or accuracy percentages should only be included when they have been measured and recorded.

12.Limitations and Future Enhancements

The usefulness of the platform depends on the quality of its feedback and the reliability of the underlying services. AI-generated suggestions may not always be perfect, so users should treat them as preparation support rather than a guarantee of interview success. The available dashboard template also does not show enough information to confirm the exact implementation of every feature.

Future work could include more detailed progress history, personalised practice based on a user's previous answers, additional interview roles and difficulty levels, downloadable performance reports, improved accessibility, and clearer explanations of how scores are calculated. The project could also be extended with more robust user profiles and secure storage of interview history, depending on the existing backend design.

13.Conclusion

InterviewGen AI is intended to make interview preparation more organised and easier to follow. It brings mock interview practice, resume review, an AI assistant, voice practice, analytics, and improvement guidance together through a central dashboard.

The project is built around a simple idea: preparation should not stop after answering a question. Users should also have a way to review their performance and decide what to practise next. With further testing and development, InterviewGen AI can continue to grow into a useful support tool for students preparing for interviews.